ITA0608 - Machine Learning
List of Lab Exercises

8.Write a program to implement Linear Regression (LR) algorithm in python.

Aim

To implement the **Linear Regression (LR)** algorithm in Python and predict the output values based on the input data.

Algorithm

1. Import the required Python libraries.
2. Load the dataset.
3. Select the input feature (X) and target (y).
4. Split the dataset into training and testing sets.
5. Create a Linear Regression model.
6. Train the model using the training data.
7. Predict the output for the test data.
8. Calculate and display the model accuracy (R^2 score).

Program Code

```
from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

# Load dataset

data = load_diabetes()

X = data.data

y = data.target


# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Create Linear Regression model
model = LinearRegression()
# Train the model
model.fit(X_train, y_train)
# Predict
y_pred = model.predict(X_test)
# Accuracy (R² Score)
score = model.score(X_test, y_test)
print("Predicted Values:")
print(y_pred[:10])
print("R2 Score:", score)
```

Output

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Load dataset
data = load_diabetes()
X = data.data
y = data.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy (R² Score)
score = model.score(X_test, y_test)

print("Predicted Values:")
print(y_pred[:10])
print("R2 Score:", score)
```

```
Predicted Values:
[139.5475584  179.51720835 134.03875572 291.41702925 123.78965872
  92.1723465  258.23238899 181.33732057  90.22411311 108.63375858]
R2 Score: 0.4526027629719195
```

Result

The **Linear Regression algorithm** was successfully implemented in Python. The model was trained on the dataset, predicted the output values, and its performance was evaluated using the **R^2 score**.